Speech Encryption Using the Nonlinear Filter Generator

A. Belmeguenai, K. Mansouri, A. Medoued

Abstract— **In this communication, we propose a stream cipher system based on nonlinear filter generator for speech data encryption. The cipher is simple and highly efficient. It is implemented to encrypt and decrypt speeches data. Several experimental tests are done for proving the system performance. The results show that the proposed scheme is secure certain attacks.**

Keywords—**Keystrem Generator, Stream Ciphers, Speech Encryption, Security evaluation.**

I. INTRODUCTION

The speech data transmissions become more and more widely used in everyday life and even have been known to be vulnerable to interception and unauthorized access. The need to protect the transfer of such data is necessary. Several papers of speech data encryption have been proposed [1], [2], [3], [4], [5]. But the characteristics of the speech data make the techniques [1] and [2] not best way for secure speech data transmissions due to the nature of speech and inherent features, especially high volume speech data. A stream cipher is very good way for secure speech data transmissions.

A stream cipher is very important part of symmetric encryption algorithm; it consists generally based on a keystream generator. The keystream generator may be a nonlinear combination generator or nonlinear filter generator in the both cases the generator takes as input a secret key and then generates a pseudorandom sequence $Z_1, Z_2, ..., Z_i$ of digits known as the running keystream.

The cipher message $Q_1, Q_2, ..., Q_i$ is obtained by the bitwise addition of the keystream digit $Z_1, Z_2, ..., Z_i$ and the plain message digit $P_1, P_2, ..., P_i$ as follows:

$$Q_i = Z_i \oplus P_i . \quad (1)$$

The receiver decrypted the cipher message by using the same keystream and adding it to the cipher message such as:

$$P_i = Z_i \oplus Q_i . \quad (2)$$

In this paper, we present a simple and secure stream cipher algorithm based on nonlinear filter generator for speech data encryption. This algorithm has the advantage of not propagating the errors [6].

II. NONLINEAR FILTER GENERATOR

We describe briefly a nonlinear filter generator. It consists on the use single linear feedback shift register (LFSR) with length L and filtering function f .

By $(s_i)_{i\geq 0}$ we denotes the sequence generated by the LFSR, the output sequence $(Z_i)_{i\geq 0}$ of the filter generator is given by

$$Z = f(s_{i+1}, s_{i+2}, ..., s_{i+m}), \text{ for every } i \geq 0 \quad (3)$$

Where f is a function of m variables, such that $m \leq L$.

The linear complexity $\Lambda(Z)$ of the keystream sequence Z computed as

$$\Lambda(Z) \leq \sum_{i=0}^{d} \binom{L}{i} \quad (4)$$

Where d denotes the algebraic degree of f . If L is a large prime, $\Lambda(Z)$ is at least equal to $\binom{L}{d}$, for more detail see [7], [8] and [9].

III. PROPOSED KEYSTREAM GENERATOR

The proposed keystream generator is based on the use one (LFSR) with large secret key $K = k_1, k_2, ..., k_{1279}$ having the primitive feedback polynomial as:

$$p(x) = x^{1279} + x^{418} + 1 \quad (5)$$

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Which is filtered by nonlinear Boolean function (called filtering function), which is a good cryptographic properties.

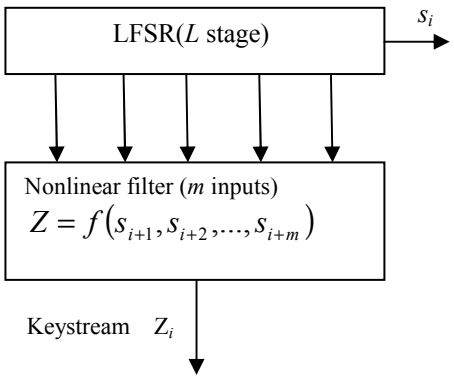


Figure 1. Nonlinear filter generator.

The output bit of LFSR at time i is computed by the recurrence relation as:

$$s_{i+1279} = s_{i+1278} \oplus s_{i+417} . \quad (6)$$

The output sequence $(s_i)_{i\geq 0}$ of LFSR has a period $2^{1279}-1$. $s_0, s_1, ...s_{1278}$ is a nonzero initial stat of LFSR.

The output sequence $(Z_i)_{i\geq 0}$ of the proposed filter generator at each time i, is given by

$$Z_i = f(\overline{X_i}) \quad (7)$$

Where f is of 14 variables function presented in [10], it is balanced, correlation immune of order 5, has algebraic degree 7 and nonlinearity $Nf = 2^{12} - 2^7$, and $\overline{X_i}$ is the fourteen inputs of filtering function f taken from the sequence $(s_i)_{i\geq 0}$ generated by the proposed LFSR.

The input $\overline{X_i}$ is selected as:

$$\overline{X_i} = \begin{pmatrix} s_{i+3}, s_{i+11}, s_{i+21}, s_{i+31}, s_{i+121}, s_{i+331}, s_{i+461}, \\ s_{i+493}, s_{i+512}, s_{i+641}, s_{i+761}, s_{i+801}, s_{i+901}, s_{i+931} \end{pmatrix} \quad (8)$$

IV. Method Description

In this section, a speech transmission security method based on nonlinear filter generator is proposed. The method is developed to encrypt and decrypt speech data as follows:
Loaded the original speech data and convert the original speech data into normalized speech data (to obtain values between 0 and 255) and convert it into normalized digital speech and store it in the file P. Then encrypt the normalized digital speech and send it to the receiver. The flow chart of the encryption process is depicted by figure 2.

The function of the receiver is to decrypt the encrypted normalized digital speech file and convert it into speech data. The flow chart of the decryption process is depicted in figure 3.

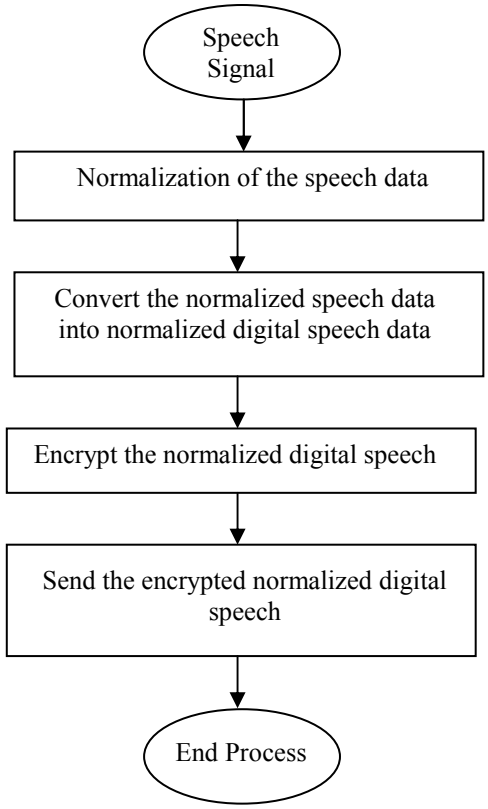


Figure 2. The flow chart of the encryption process.

A. Keystream Algorithm

- *Load the bits of initialization of LFSR* $s_1, s_2, ..., s_{1279}$;
- *Generate the binary sequence* s_i, *the output of LFSR* using $s_{i+1279} = s_{i+1278} \oplus s_{i+417}$;
- *Calculate the keystream* Z_i *generated by the function* f *as* $Z_i = f(\overline{X_i})$;

V. Test and Experimental Results

Some experimental results are given in this section to demonstrate the efficiency of the proposed cipher and also we discuss the obtained results from implementing the proposed cipher. In order to implement such a system, one must go through several steps which were described in details in the preceding sections.

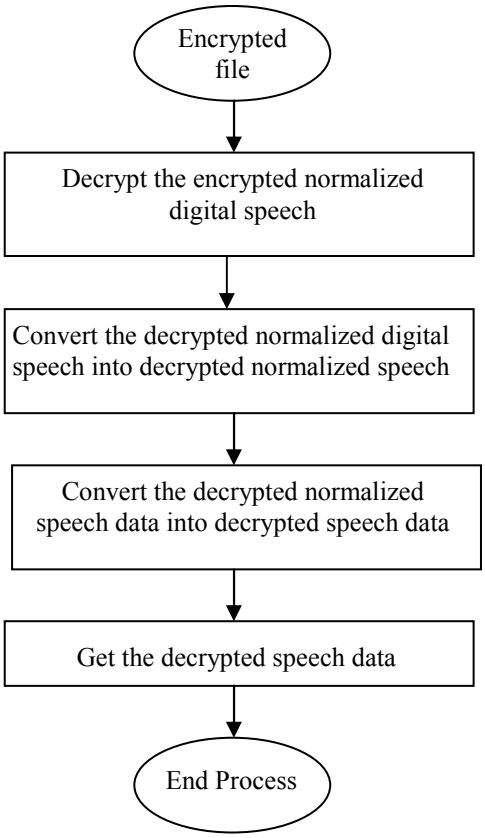
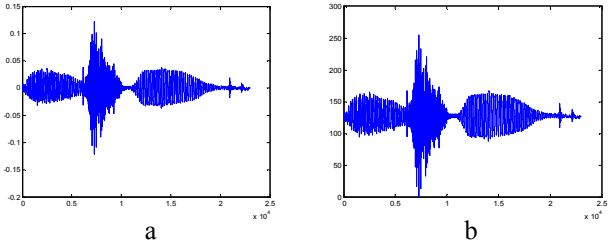


Figure 3. The flow chart of the decryption process.

In this communication three speech data depicted in figures 4, 5 and 6 were recorded from different speakers and were saved as .wav file format. To encrypt these speech data, first we convert the original speech data .wav file into normalized speech data, and then convert the normalized speech data into normalized digital speech file. The encryption programs take the normalized digital speech file as input and performed the encryption operation on the file to produce an unreadable encrypted normalized digital speech file and saved it on another file.

On the other hand, the decryption program take the encrypted normalized digital speech file as input and performed decryption operation on the file to produce the decrypt normalized digital speech file, then go through several steps which were described in decryption process to get the original speech data. Figure 4, 5 and 6 shows the encryption and decryption results.



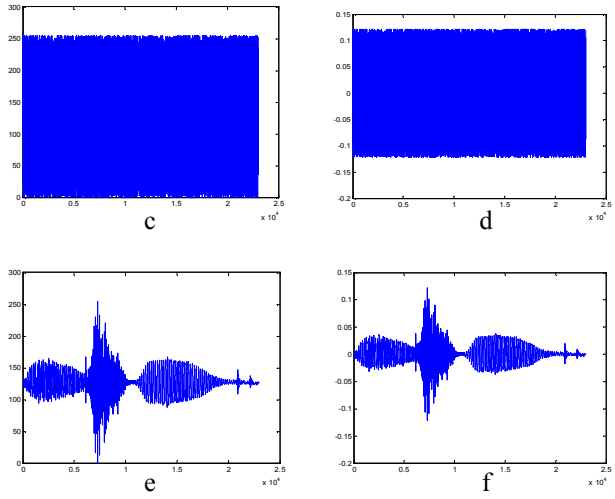


Figure 4. Test for Speech 1: (a) original speech, (b) normalized speech, (c) encrypted normalized speech, (d) encrypted speech of original speech, (e) decrypted normalized speech, (f) decrypted speech.

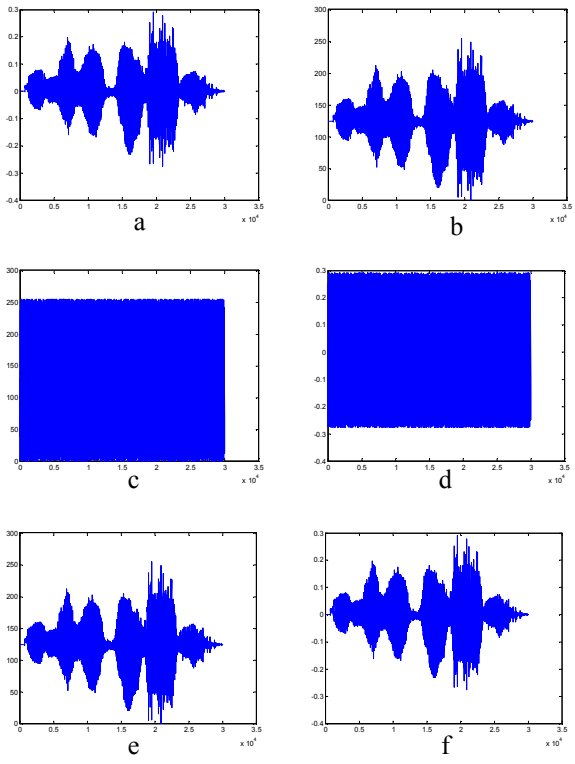
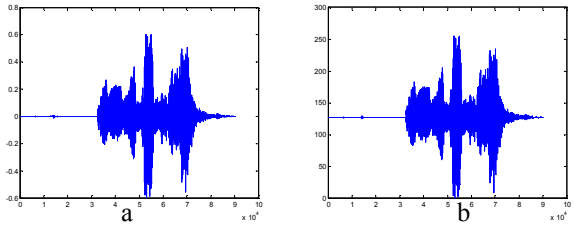


Figure 5. Test for Speech 2: (a) original speech, (b) normalized speech, (c) encrypted normalized speech, (d) encrypted speech of original speech, (e) decrypted normalized speech, (f) decrypted speech.



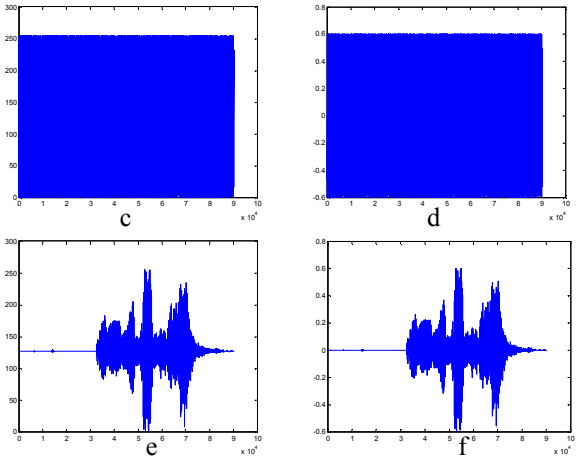


Figure 6. est for Speech 3: (a) original speech, (b) normalized speech, (c) encrypted normalized speech, (d) encrypted speech of original speech, (e) decrypted normalized speech, (f) decrypted speech.

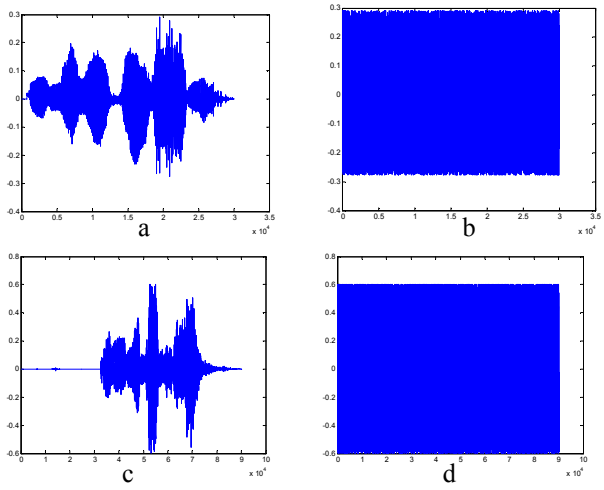


Figure 7. Sensitivity analysis: (a) original speech 2 , (b) show their decrypted speech by incorrect key, (c) original speech 3 , (d) show their decrypted speech by incorrect key.

VI. SECURITY EVALUATION

To prove the robustness of the proposed cipher, we discuss the security evaluation of the proposed scheme.

A. *Correlation Coefficient Analysis*

Any speech encryption system is said to be good, if encryption algorithm hides all attributes of an original speech, and encrypted speech is totally random and highly uncorrelated. Table 1 shows the correlation coefficient results.

In Table 1 we denote by Cor1, Cor2 and Cor3 respectively the correlation coefficient between: original speech and its encrypted speech, original speech and its decrypted speech by incorrect key, and original speech and its decrypted speech.

TABLE I. CORRELATION COEFFICIENTS ANALYSIS

Speech	Cor_1	Cor_2	Cor_3
Speech 1	-0,0003564	0.0026	1
Speech 2	0,00035654	-0.0032	1
Speech 3	-0.0024	-0.0047	1

B. *Key Sensitivity*

The ideal speech encryption process should be sensitive with the secret key. It means that the change of a one bit in the secret key cannot decrypt a decrypted speech. So, we have decrypted the encrypted speech 2 and 3 depicted in figure 5 and 6 by different key (one bit changed). Figure 7 gives the sensitivity of the proposed encryption system.

C. *Key Space Analysis*

To prevent some attacks like brute force, the key spaces should be large. So an encryption cipher considered secure if its key space is large enough. In proposed algorithm the key space K can be equal to K = 1279 bits, the attacker needs 2^{1279} tests by exhaustive research. This is very large because an attacker needs to try all possible keys. So the proposed cipher is free from the brute force attack.

D. *Berlekamps-Massey attack*

In order to resist a Berlekamp-Massey algorithm, the linear complexity $\Lambda(Z)$ of the keystream sequence $(Z_i)_{i\geq 0}$ must highest possible, because the Berlekamps-Massey algorithm [11] requires $2\Lambda(Z)$ successive data to work. So the linear complexity $\Lambda(Z)$ has at least equal to 2^{60}, this bound make ineffective Berlekamp-Massey algorithm.

VII. CONCLUSION

This work presents an approach base on the nonlinear filter generator for speech data encryption and decryption. Three different speeches are used for validate the proposed speech cipher. To prove the proposed scheme is resist to some attacks, a series of test are given like key sensitivity test, key space test, and Berlekamp-Massey algorithm test and correlation coefficient test. The results show that the proposed speech cipher provides an efficient and secure way for speech encryption and transmission.

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